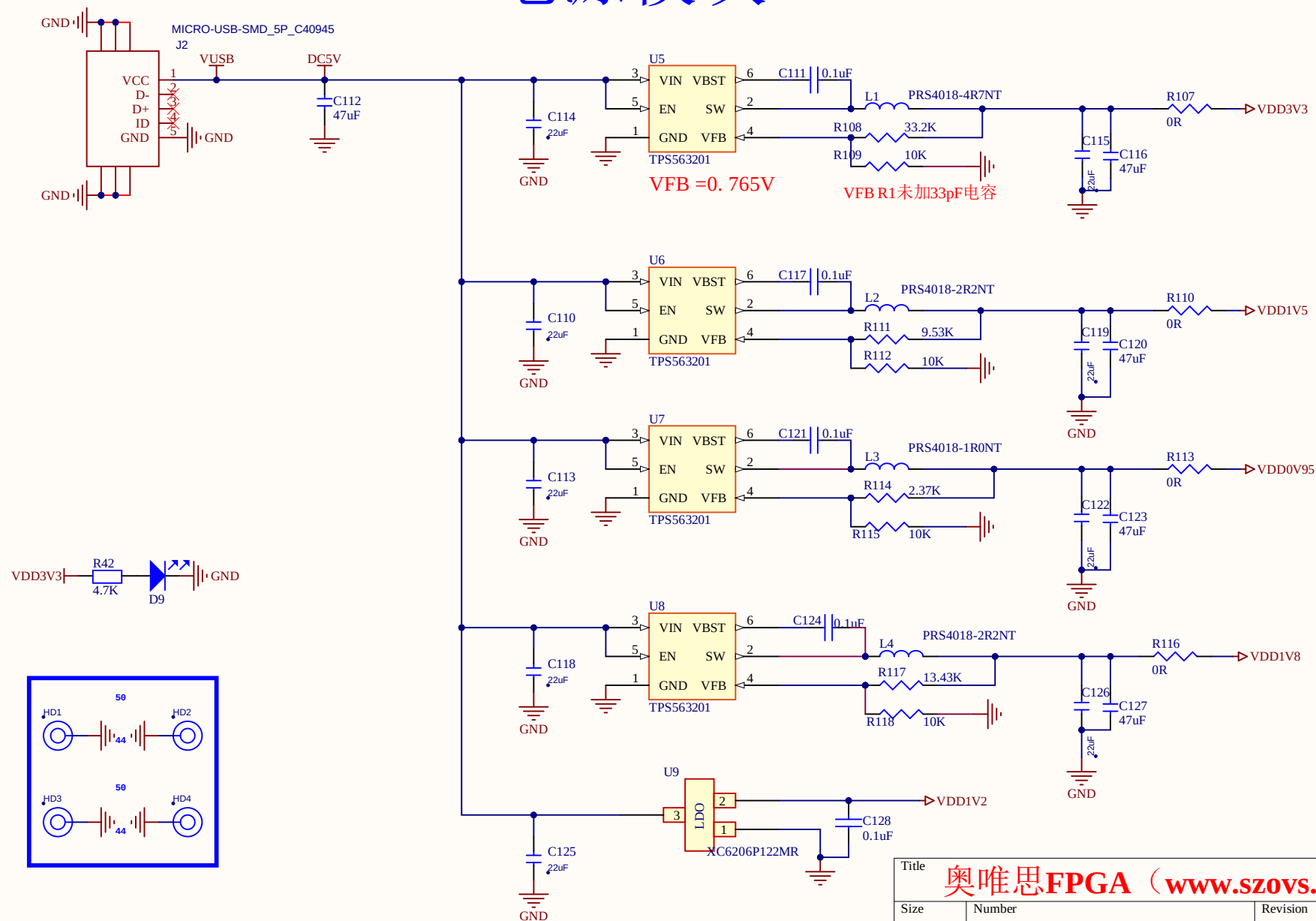
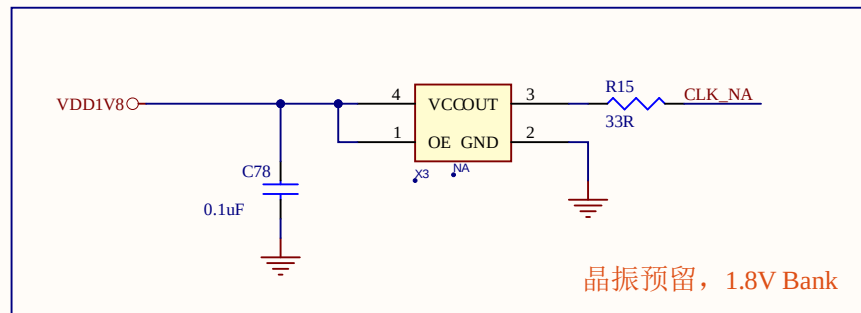
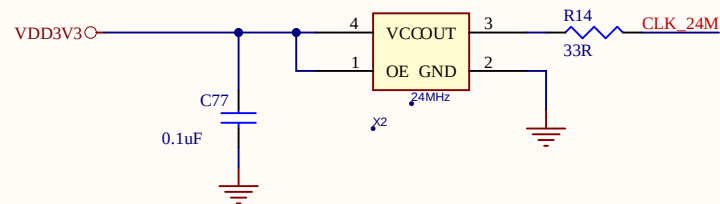
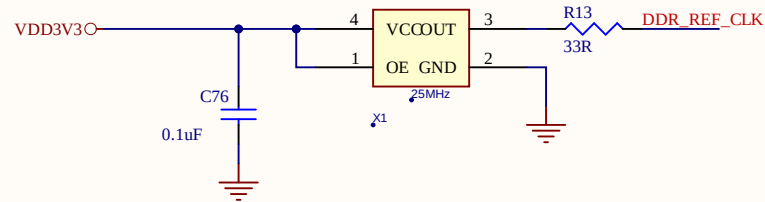


FPGA 电源模块



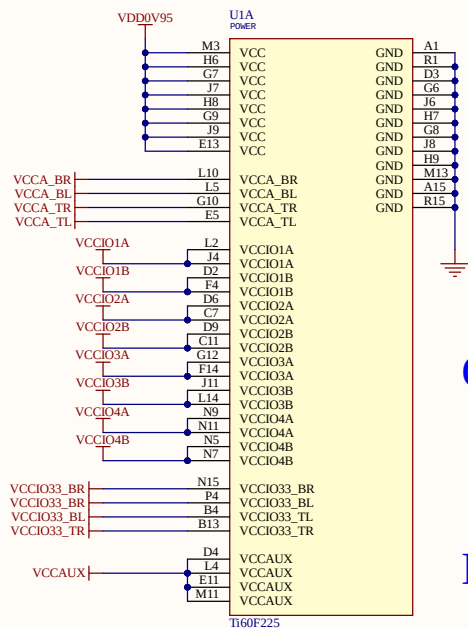
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Size A4	Number	Revision
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FPGA 板载晶振



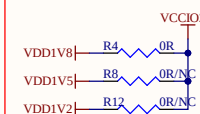
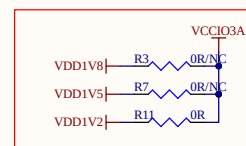
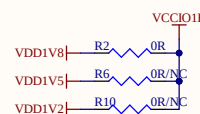
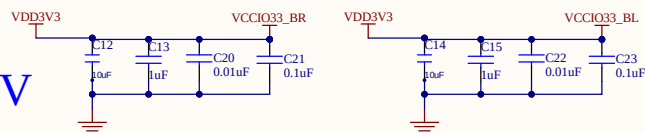
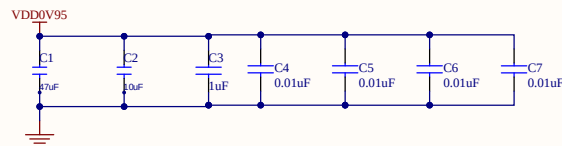
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Ti60F225 Power

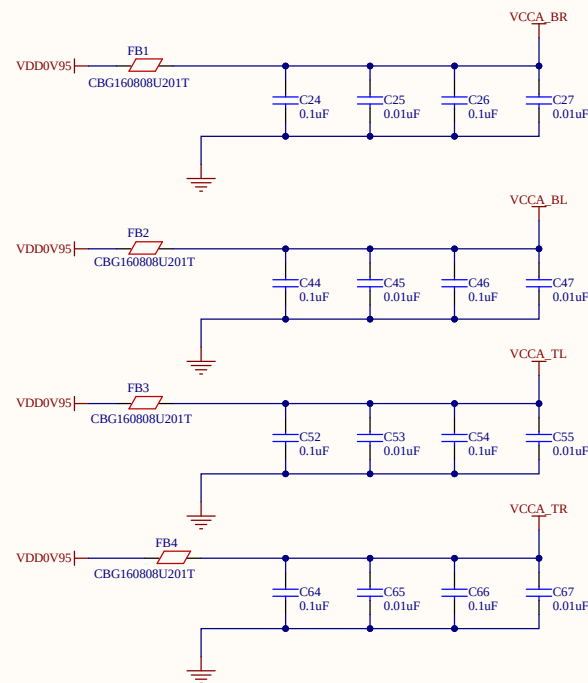
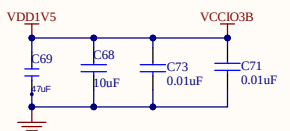
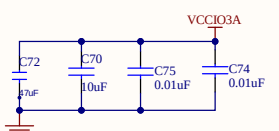
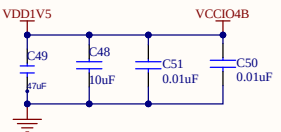
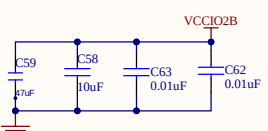
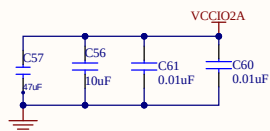
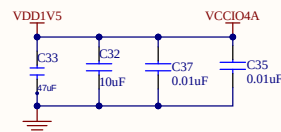
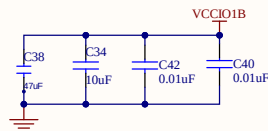
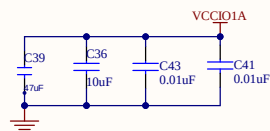
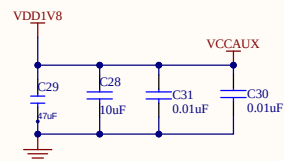


GPIO = 3.3V

Default 1.8V



BANK-3A 需1.2V 走MIPI信号



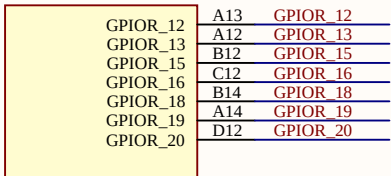
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Size A3	Number	Revision
Date: 9/14/2023	Sheet of	
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BANK 3.3V GPIO

共21个3.3V GPIO

U1H

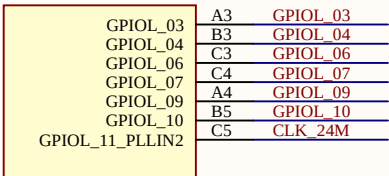
TR



Ti60F225

U1E

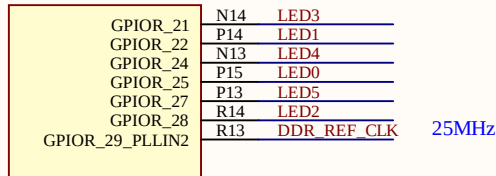
TL



Ti60F225

U1K

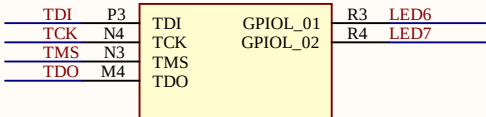
BR



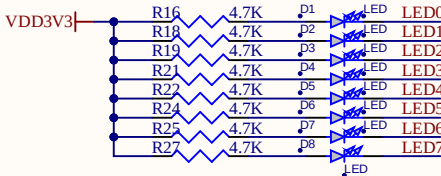
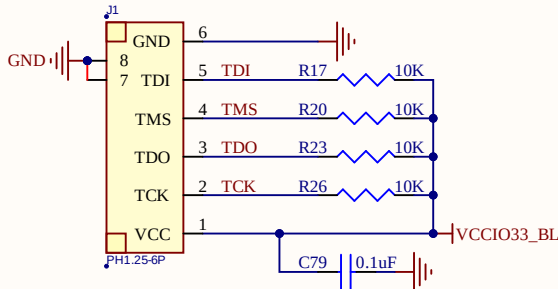
Ti60F225

U1B

BL



Ti60F225



Title 奥唯思FPGA (www.szovs.com)		
Size A4	Number	Revision
Date: 9/14/2023	Sheet of 4	
File: D:\PCB_Project_V2\04_BANK_3V3_Schematic	Drawn By:	

DDR3模块

DDR3 SDRAM A (16M X 16 X 8)

U1J
BANK_3B

GPIOR_N_09_CLK10_N H15 DDR3_CLK_N
GPIOR_P_09_CLK10_P J15 DDR3_CLK_P
GPIOR_N_08_CLK11_N H11 DDR3_RST_N
GPIOR_P_08_CLK11_P H10 DDR3_A10
GPIOR_N_07 GPIOR_N_07
GPIOR_P_07 GPIOR_P_07
GPIOR_N_06_CDI21 GPIOR_N_06_CDI21
GPIOR_P_06_CDI20 GPIOR_P_06_CDI20
GPIOR_N_05_CDI31 GPIOR_N_05_CDI31
GPIOR_P_05_CDI30 GPIOR_P_05_CDI30
GPIOR_N_04_CDI29 GPIOR_N_04_CDI29
GPIOR_P_04_CDI28 GPIOR_P_04_CDI28
GPIOR_N_03_CDI27 GPIOR_N_03_CDI27
GPIOR_P_03_CDI26 GPIOR_P_03_CDI26
GPIOR_N_02_CDI25 GPIOR_N_02_CDI25
GPIOR_P_02_CDI24 GPIOR_P_02_CDI24
GPIOR_N_01_CDI23 GPIOR_N_01_CDI23
GPIOR_P_01_EXTFB GPIOR_P_01_EXTFB
GPIOR_N_00_CDI22 GPIOR_N_00_CDI22
GPIOR_P_00_PLIN0 GPIOR_P_00_PLIN0

T160F225

R77

10K

U1L
BANK_4A

GPIOB_N_17 P12 DDR3_D4
GPIOB_P_17_PLLIN1 P11 DDR3_D7
GPIOB_N_15_CDI19 R11 DDR3_DQS_N0
GPIOB_P_15_CDI18 R12 DDR3_DQS_P0
GPIOB_N_14_CDI17 P10 DDR3_D6
GPIOB_P_14_CDI16 R10 DDR3_D2
GPIOB_N_13_CDI15 M10 DDR3_D1
GPIOB_P_13_CDI14 P9 DDR3_D3
GPIOB_N_12_CDI13 R9 DDR3_D0
GPIOB_P_12_CDI12 K9 DDR3_A15
GPIOB_N_11_CDI11 K8 DDR3_A5
GPIOB_P_11_CDI10 L9 DDR3_BA1
GPIOB_N_10_CLK12_N M9 DDR3_DM0
GPIOB_P_10_CLK12_P L8 DDR3_CKE
GPIOB_N_09_CLK13_P K7
GPIOB_P_09_CLK13_P

T160F225

R86

10K

U1M
BANK_4B

GPIOB_N_08_CLK14_N M8 DDR3_CS_N
GPIOB_P_08_CLK14_P N8 DDR3_D11
GPIOB_N_07_CLK15_N P8 DDR3_DM1
GPIOB_P_07_CLK15_P L7 DDR3_BA0
GPIOB_N_06_CDI9 M7 DDR3_ODT
GPIOB_P_06_CDI8 K6 DDR3_A3
GPIOB_N_04 GPIOB_N_04
GPIOB_P_04_SSU_N M6 DDR3_D9
GPIOB_N_03_CDI7 N6 DDR3_D13
GPIOB_P_03_CDI6 P7 DDR3_DQS_N1
GPIOB_N_02_CDI5 R7 DDR3_DQS_P1
GPIOB_P_02_CDI4 P6 DDR3_D12
GPIOB_N_01_EXTFB R6 DDR3_D10
GPIOB_P_01_EXTFB P5 DDR3_D15
GPIOB_N_00_PLIN1 R5 DDR3_D14
GPIOB_P_00_PLIN1

T160F225

REF_RES_4B

R120

10K

DDR3 Device
A0 DQ0
A1 DQ1
A2 DQ2
A3 DQ3
A4 DQ4
A5 DQ5
A6 DQ6
A7 DQ7
A8 DQ8
A9 DQ9
A10/AP DQ10
A11 DQ11
A12/BCn DQ12
A13 DQ13
A14 DQ14
A15 DQ15

DDR3_CKE K9 CKE
DDR3_CLK_P J7 CK_P
DDR3_CLK_N K6 CK_N

DDR3_CS_N L2 CS
DDR3_WE_N L3 WE
DDR3_RAS_N J3 RAS
DDR3_CAS_N K3 CAS

DDR3_BA0 M2 BA0
DDR3_BA1 M3 BA1
DDR3_RST_N T2 RESETn
DDR3_ODT K1 ODT

H1 HREFDQ
M8 VREFCA
B1 VSSQ
B9 VSSQ
D1 VSSQ
D8 VSSQ
E8 VSSQ
F9 VSSQ
G9 VSSQ
N1 VDD
N9 VDD
R1 VDD
R9 VDD
A1 VDDQ
A8 VDDQ
C1 VDDQ
C9 VDDQ
D2 VDDQ
E9 VDDQ
F1 VDDQ
H2 VDDQ
H9 VDDQ

VREF_DDR3
VDD1V5
VDD3V3
VDD1V5

CAD Note:
Place decoupling caps
near DDR3 SDRAM A power
pins

C90 0.1uF
C91 0.01uF
C92 1nF
C93 1nF
C94 1nF
C95 1nF
C96 1uF
C97 1uF
C98 0.1uF
C99 0.1uF
C100 0.1uF
C101 0.01uF
C102 0.01uF
C103 0.01uF

DDR3_CLK_P
DDR3_CLK_N
R52 100R

DDR3_VTT
R55 51R

DDR3_CKE
R59 4.7K

DDR3_VTT
R83 1uF
R84 0.1uF
R85 0.1uF
R86 0.1uF
R87 0.1uF
R88 0.1uF
R89 0.1uF

VDD3V3
C129 4.7uF
VDD1V5
C130 47uF

VDD3V3
VDD1V5
U10
VDD
S3
S5
VDDQSN
VLDON
VTT
VTTNS
VTTREF

TPSS1206
WSON(2X2)

0402
0805
C131 47uF
C132 47uF
C133 47uF
C134 4.7uF

0402
0805
C135 4.7uF

VDD1V5
R82 100R/0.01%
R83 100R/0.01%
R84 100R/0.01%
R85 100R/0.01%

0.1uF
C104
C106
C107
C108

0.1uF
0.1uF
0.1uF
0.1uF

0.1uF
0.1uF
0.1uF
0.1uF

0.1uF
0.1uF
0.1uF
0.1uF

DDR3_D0 R28 33R
DDR3_D1 R29 33R
DDR3_D2 R30 33R
DDR3_D3 R31 33R

DDR3_D4 R32 33R
DDR3_D5 R33 33R
DDR3_D6 R34 33R
DDR3_D7 R35 33R

DDR3_D8 R36 33R
DDR3_D9 R37 33R
DDR3_D10 R38 33R
DDR3_D11 R39 33R

DDR3_D12 R40 33R
DDR3_D13 R41 33R
DDR3_D14 R42 33R
DDR3_D15 R43 33R

DDR3_DQS_P0 R44 33R
DDR3_DQS_N0 R45 33R
DDR3_DQS_P1 R46 33R
DDR3_DQS_N1 R47 33R

DDR3_DM0 R48 33R
DDR3_DM1 R49 33R

DDR3_VTT
R54 51R
R55 51R
R56 51R
R57 51R
R58 51R
R59 51R
R60 51R
R61 51R
R62 51R
R63 51R
R64 51R

DDR3_CS_N R65 51R
DDR3_ODT R66 51R
DDR3_BA2 R67 51R
DDR3_A3 R68 51R
DDR3_BA0 R69 51R
DDR3_A10 R70 51R
DDR3_RAS_N R71 51R
DDR3_CAS_N R72 51R

DDR3_WE_N R73 51R
DDR3_A6 R74 51R
DDR3_A13 R75 51R
DDR3_A11 R76 51R
DDR3_A1 R77 51R
DDR3_A6 R78 51R
DDR3_A4 R79 51R
DDR3_BA1 R80 51R
DDR3_BA1 R81 51R

DDR3_VTT
R82 100R/0.01%
R83 100R/0.01%
R84 100R/0.01%
R85 100R/0.01%

0.1uF
C104
C106
C107
C108

0.1uF
0.1uF
0.1uF
0.1uF

0.1uF
0.1uF
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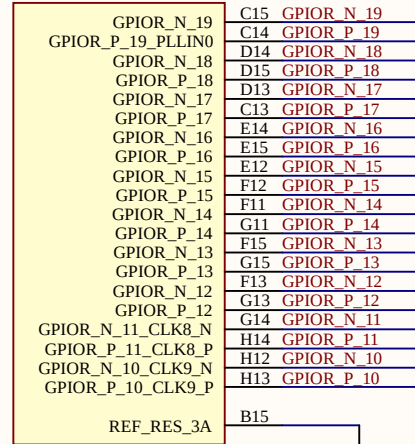
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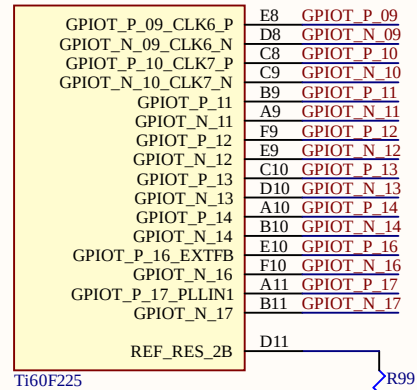
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FPGA核心板 P1 B2B接口

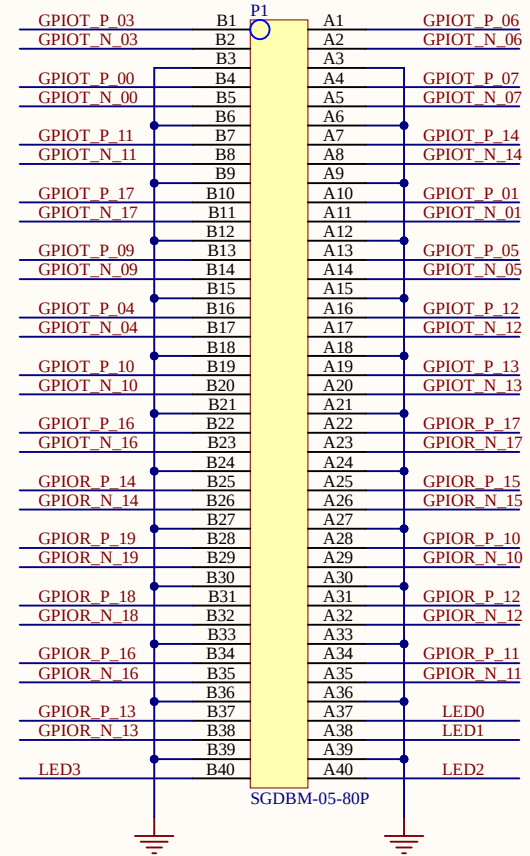
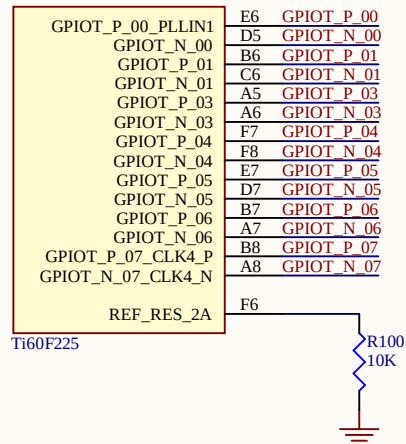
U1I
BANK_3A



U1G
BANK_2B



U1F
BANK_2A



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Size A4	Number	Revision
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File: D:\PCB_Project_V2\..06_P1_Conector_Schematic	Drawn By:	

FPGA核心板 P2接口相关

U1D
BANK_1B

GPIOL_P_09_CLK2_P H1 GPIOL P 09
GPIOL_N_09_CLK2_N G1 GPIOL N 09
GPIOL_P_10_CLK3_P H2 GPIOL P 10
GPIOL_N_10_CLK3_N G2 GPIOL N 10
GPIOL_P_11 F5 GPIOL P 11
GPIOL_N_11 F5 GPIOL N 11
GPIOL_P_12 G3 GPIOL P 12
GPIOL_N_12 G3 GPIOL N 12
GPIOL_P_13_CBSEL0 E3 GPIOL P 13 CBSEL0
GPIOL_N_13_CBSEL1 E3 GPIOL N 13 CBSEL1
GPIOL_P_14 F1 GPIOL P 14
GPIOL_N_14 F1 GPIOL N 14
GPIOL_P_15_NSTATUS E1 GPIOL P 15 NSTATUS
GPIOL_N_15_TEST_N E2 GPIOL N 15 TEST_N
GPIOL_P_16 D1 GPIOL P 16
GPIOL_N_16 C1 GPIOL N 16
GPIOL_P_17_EXTFB B1 GPIOL P 17
GPIOL_N_17 C2 GPIOL N 17
GPIOL_P_18_PLLIN0 B2 GPIOL P 18
GPIOL_N_18 A2 GPIOL N 18

REF_RES_1B

Ti60F225

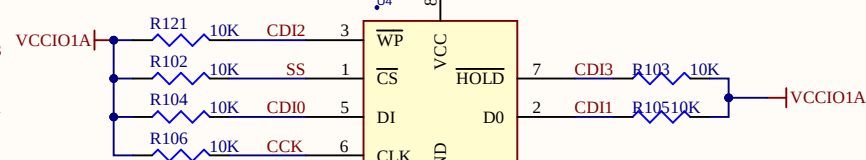
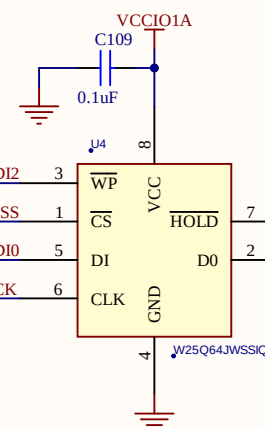
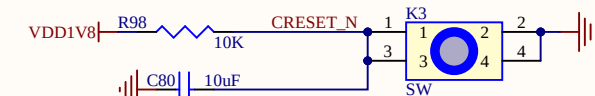
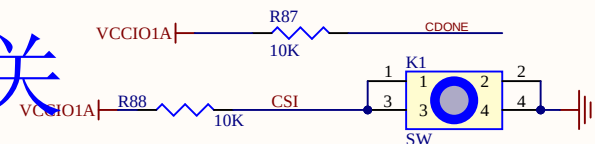
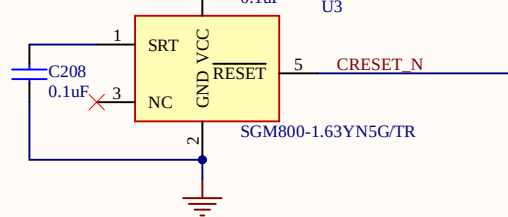
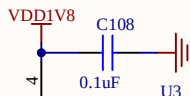
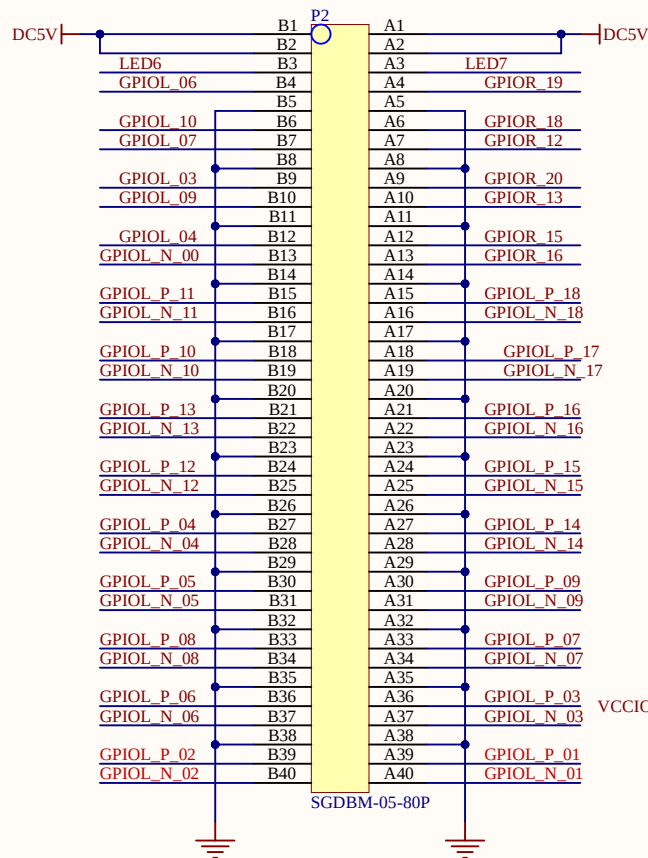
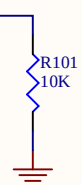


U1C
BANK_1A

CDONE L3 CDONE
CRESET_N J5 CRESET_N
GPIOL_P_00_PLLIN0 P2 CLK NA
GPIOL_N_00 R2 GPIOL N 00
GPIOL_P_01_SSL_N P1 GPIOL P 01 SS
GPIOL_N_01_CCK N1 GPIOL N 01 CCK
GPIOL_P_02_CSI N2 GPIOL P 02 CSI
GPIOL_N_02_CS0 M2 GPIOL N 02 CS0
GPIOL_P_03_CDI0 M1 GPIOL P 03 CDI0
GPIOL_N_03_CDI1 L1 GPIOL N 03 CDI1
GPIOL_P_04_CDI2 H5 GPIOL P 04 CDI2
GPIOL_N_04_CDI3 H4 GPIOL N 04 CDI3
GPIOL_P_05 K4 GPIOL P 05
GPIOL_N_05 J3 GPIOL N 05
GPIOL_P_06 K3 GPIOL P 06
GPIOL_N_06 K2 GPIOL N 06
GPIOL_P_07_CLK0_P K1 GPIOL P 07
GPIOL_N_07_CLK0_N J1 GPIOL N 07
GPIOL_P_08_CLK1_P J2 GPIOL P 08
GPIOL_N_08_CLK1_N H3 GPIOL N 08

REF_RES_1A

Ti60F225



奥唯思FPGA (www.szovs.com)

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A4		
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File:	D:\PCB_Project_V2\107_P2_Conector_Schematic	Drawn By: